

Comprehensive Frame-by-Frame Analysis — NASA Artemis II Re-entry Broadcast

Source: 266 screenshot frames extracted from “NASA’s Artemis II Crew Comes Home (Official Broadcast)” (YouTube [watch?v=nfhDu0Hmp0A](https://www.youtube.com/watch?v=nfhDu0Hmp0A))

Scope: On-screen telemetry dials during the re-entry / plasma-blackout segment (video time $\approx$ 1:28:19 $\rightarrow$ 1:31:33)

Method: OCR of each frame’s three dials — **VELOCITY M/H, FROM EARTH Miles, TO MOON Miles** — followed by **manual visual verification** of every value flagged as anomalous.

This is a revised report. An earlier version of this analysis contained several errors caused by OCR misreads and a misunderstanding of the on-screen time field. Those errors, and the corrections, are documented in full in Section 2 below. This revision separates **what is provable from the frames** from **what is assumption or interpretation**, per the project’s accuracy standard.

1. What the footage actually shows (critical context)

Before any anomaly claim, the nature of the footage must be stated plainly, because it changes how the dial readings should be interpreted:

- The segment is **explicitly labeled “VISUALIZATION”** on-screen (top-left of the video frame).
- The broadcast narration during this segment states, in substance, “this is a visualization of the plasma build-up around the spacecraft.”
- This segment coincides with the **communications blackout** during peak heating. The narration explicitly notes NASA does **not** have live communications from the crew during this window (e.g., “still do not have communications from the crew”).

Implication: The three dials in this segment are part of an **acknowledged animation/model**, shown during a period when live hardware telemetry is not being received. They therefore represent **modeled/predicted values driving a graphic**, not a live downlink from the vehicle. Any irregularity in these dials is an irregularity **in a visualization**, and must be described as such — not as a discrepancy in NASA’s flight telemetry.

2. Corrections to the prior analysis

The following claims appeared in the earlier report. Each has been re-examined against the original frames and is **retracted or corrected**:

#	Prior claim	Finding on re-check	Status
1	"-142 minute time reset at scene 2817 (jumps to 3:53:40)"	The video overlay shows time as CURRENT / TOTAL . "3:53:40" is the total video duration , not a timestamp. Scene 2817's actual current time is 1:31:25 , fully consistent with its neighbors (2806 = 1:31:24, 2828 = 1:31:26). The OCR had captured the duration field.	RETRACTED (false positive)
2	" +2,937 mph velocity reversal at scenes 2872 / 2883"	OCR misread the leading digit 1 as 4 : "11,778" → "14,778" and "11,745" → "14,745". The true sequence is a smooth deceleration 11,841 → 11,778 → 11,745 → 11,711.	RETRACTED (OCR error)
3	Scene 144 velocity "25,108"	Correct value on visual check is 25,105 .	CORRECTED
4	Scene 265 To-Moon "248,256"	Correct value on visual check is 248,254 .	CORRECTED

Process note: Two independent OCR passes produced the same digit-misread in case #2, so OCR cross-checking alone was insufficient — every flagged value was ultimately confirmed by reading the pixels of the original frame crop. The corrected values are reflected in `telemetry_clean.json` and `complete_telemetry_table.*`.

Two interpretive claims from the prior report are also withdrawn as not substantiated:

- "Velocity increasing during re-entry is anomalous." A small climb (25,064 → 25,192 mph over ~25 s) before/at atmospheric entry interface is consistent with continued gravitational acceleration prior to significant atmospheric drag. It is not, on its own, evidence of an error.
 - "Geometrically impossible that both Earth-distance and Moon-distance increase." Both distances increasing simultaneously is geometrically possible during an altitude-gaining (skip) phase and is not a contradiction.
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3. Verified observations (provable from the frames)

These are stated conservatively and are directly checkable in the data table.

3.1 Byte-identical (duplicate) frames

Of the 266 frames, **only 155 are unique**; 111 are exact byte-for-byte duplicates (verified by MD5 hash) of another frame. Three duplicate blocks account for the bulk:

Block	Frame hash	# frames	Scene range	Video time shown	Dials shown
A	318cbfe6	65	430-1134	1:28:20	v=25,067 · earth=28 · moon=248,177
B	c6ca2230	47	1-1673	1:28:19	v=25,064 · earth=28 · moon=248,174
C	f5accc2c	2	419-1167	1:28:19	v=25,064 · earth=28 · moon=248,175

Honest interpretation: Every frame within a duplicate block **shares the same on-screen video time**. They are therefore the same broadcast moment captured multiple times by the screenshot process — **not** different points in time showing frozen numbers. This is the single most important correction to the project's working hypothesis (see Section 5). It is consistent with the segment being a held/looping visualization graphic during the blackout.

3.2 The 110-second jump

Between the end of the duplicated 1:28:19-1:28:20 material (scene 1673) and the next distinct frame (scene 1684, time **1:30:09**), the video time advances ~110 s and velocity drops **25,064 → 14,492 mph**. This is the peak-heating / blackout interval. The dials resume smooth behavior after it. Whether this represents a genuine ~110 s hold of the graphic or simply a gap in the screenshot sampling **cannot be determined from the frames alone** and is left as undetermined.

3.3 Minor velocity non-monotonicities in the second segment

After the jump, velocity generally decelerates ($\approx 14,492 \rightarrow 11,650$ mph) but shows six small increases between consecutive advancing frames:

Between scenes	Video time	Velocity change
2069 → 2080	1:30:34→35	+30 mph
2102 → 2113	1:30:36→37	+61 mph
2146 → 2157	1:30:39→40	+101 mph
2245 → 2256	1:30:46→47	+166 mph
2311 → 2322	1:30:50→51	+103 mph
2355 → 2366	1:30:53→54	+95 mph

These were visually confirmed as real (not OCR errors). They are small relative to the overall ~3,000 mph decline across the same window and are **consistent with interpolation/animation artifacts in a visualization graphic**. They are the only genuine non-physical micro-fluctuations found in the corrected dataset.

3.4 Overall telemetry is internally consistent and physically plausible (after correction)

- **Velocity:** monotonic-with-minor-jitter deceleration from ~25,192 mph (entry interface) down to ~11,650 mph.
- **From Earth (mi):** 28 → 32 (segment 1), then 40 → 34 (segment 2) — a dip-then-climb profile consistent with an Orion skip-entry trajectory.
- **To Moon (mi):** steady increase 248,174 → 248,628 (spacecraft moving away from the Moon).
- **Outlier scan of the corrected data: zero** remaining velocity OCR outliers; the only large velocity change is the expected drop across the 110 s blackout gap.

4. Claims NOT supported by the evidence

To keep provable findings separate from speculation, the following are explicitly **not** supported by these frames:

- No evidence that the telemetry “stayed the same while time advanced.” Every duplicate frame shares the same displayed time.
- No evidence of a time reversal, a velocity reversal, or a geometric impossibility (all three were OCR/interpretation errors — Section 2).
- No basis to characterize the modeled dial values as falsified data; they are a labeled visualization shown during the comms blackout.

5. Honest conclusion

The corrected, verified dataset does **not** support the hypothesis that “the speed should have changed but the telemetry did not.” The opposite is true in this footage: **whenever the on-screen video time advances, the telemetry changes accordingly**, and every apparently “frozen” frame is a

byte-identical duplicate that also carries the **same** displayed timestamp — i.e., the same broadcast moment captured more than once, not a later moment with stale numbers.

The genuine, defensible observations are narrower:

1. The segment is a **NASA-labeled visualization shown during the communications blackout**, so its dials are modeled values, not a live downlink.
2. The screenshot set is **~42% exact duplicates**, which can create a misleading impression of “frozen” telemetry if timestamps are not checked.
3. There are **small (≤ 166 mph) non-monotonic velocity blips** in the second segment, consistent with animation interpolation, not physics.

The earlier “anomalies” (time reset, velocity reversal, geometric impossibility) were **artifacts of OCR misreads and a misread time field**, and are withdrawn. This revision documents them transparently rather than leaving them in the record.

Data: `telemetry_clean.json` (corrected) · `complete_telemetry_table.{csv,md,txt}` · 266 source frames in `frames/`. Values obtained by OCR with manual pixel-level verification of all flagged readings. Licensed CC BY-NC 4.0.